Get ready for Cloud and Jenkins

From "it works on my machine" to "Jenkins checked it for me."

Complete this handout before the lecture. Everything you need for preparation is here: download links, commands, setup steps, and checks. No project files, class ZIPs, or separate guide are needed. The activity will be introduced during the session; no previous Jenkins or cloud experience is required.

1. Bring your laptop, charger, and a little curiosity

Allow 30-60+ minutes before class for downloads and first-time setup. Suggested lab allowance: 8 GB RAM and at least 2 GB free disk (not vendor minimum requirements). Internet is needed for initial downloads and Jenkins plugins; the prepared demo runs locally.

2. Have these tools ready

Tool	What to prepare
Java	Full 64-bit JDK 21, including java, javac, and jar. A JRE or an IDE alone is not enough.
Git	Command-line Git, including git daemon. Keep your existing installation if it works.
Editor + browser	Your existing IDE or text editor and a desktop browser. A C++-only IDE is fine for editing; install the JDK separately.
Terminal	Windows: PowerShell 5.1+. macOS/Linux: Bash, curl, and shasum (macOS) or sha256sum (Linux).
Jenkins	Download the WAR file and follow page 2. No Windows service, Docker, or separate application server is needed.

3. Official downloads

JDK 21 (Eclipse Temurin): https://adoptium.net/temurin/releases/?version=21

Git (only if missing): https://git-scm.com/downloads

Choose the JDK installer for your operating system and CPU: x64 or ARM64. On Windows, enable the installer options for PATH and JAVA_HOME, then open a new terminal. JAVA_HOME should point to the JDK folder, not its bin folder.

On macOS, use the JDK 21 PKG installer. On Linux, select JDK 21 in the distribution-specific installation instructions below. If you already have Java, keep it only if the terminal checks below report 21.x.

JDK installation help: https://adoptium.net/installation/

4. Verify Java and Git in a new terminal

```
java -version
javac -version
jar --version
git --version
```

Ready means: java, javac, and jar all report 21.x; Git reports a version. An IDE's bundled Java is not a substitute for these terminal checks.

You do NOT need Maven, Gradle, Docker, Python, a database, a GitHub account, a cloud subscription, or a payment card.

Download and start Jenkins

Use the commands for your operating system only. These commands do not require any instructor-provided scripts. Use your normal user account.

5. Create a preparation folder

Windows PowerShell:

```
New-Item -ItemType Directory -Force "$HOME\CloudJenkinsLab\jenkins-lab\downloads"
Set-Location "$HOME\CloudJenkinsLab\jenkins-lab"
```

macOS/Linux terminal:

```
mkdir -p "$HOME/CloudJenkinsLab/jenkins-lab/downloads"
cd "$HOME/CloudJenkinsLab/jenkins-lab"
```

6. Download the class version

Jenkins 2.580.1 WAR: https://get.jenkins.io/war-stable/2.580.1/jenkins.war

In your browser, save the file as jenkins.war in the downloads folder you just created. Your preparation folder is CloudJenkinsLab inside your user home folder. Do not unzip the WAR, install an MSI, or substitute another version.

Official SHA-256 checksum: https://get.jenkins.io/war-stable/2.580.1/jenkins.war.sha256

Compare the hash from the command below with the official checksum (ignore letter case). Stop and download again if they do not match.

```
Windows:      Get-FileHash .\downloads\jenkins.war -Algorithm SHA256
macOS:        shasum -a 256 downloads/jenkins.war
Linux:        sha256sum downloads/jenkins.war
```

7. Start Jenkins locally

Keep the terminal in jenkins-lab. Windows PowerShell:

```
$env:JENKINS_HOME = "$PWD\runtime\jenkins-home"
java -Xmx768m -jar .\downloads\jenkins.war --httpListenAddress=127.0.0.1 --httpPort=8080
```

macOS/Linux:

```
export JENKINS_HOME="$PWD/runtime/jenkins-home"
java -Xmx768m -jar downloads/jenkins.war --httpListenAddress=127.0.0.1 --httpPort=8080
```

Leave this terminal open while using Jenkins. When startup finishes, open http://127.0.0.1:8080 in your browser. The server is restricted to your laptop. Do not add firewall exceptions, port forwarding, or public tunnels.

Official Jenkins WAR documentation: https://www.jenkins.io/doc/book/installing/war-file/

Finish setup and check readiness

8. Complete the first-time setup wizard

1. On the Unlock Jenkins page, enter the initial administrator password shown in your startup terminal. If you cannot find it, open the file runtime/jenkins-home/secrets/initialAdminPassword inside your jenkins-lab folder in a text editor. This is a private credential.

2. Choose Install suggested plugins and wait for installation to finish. Keep internet access available. If installation fails, resolve the network problem rather than skipping required plugins.

3. Create your own administrator account and keep the password private. Set the Jenkins URL to http://127.0.0.1:8080/ if prompted, then finish setup.

4. Open Manage Jenkins > Plugins > Installed plugins and check for Git (git), Pipeline (workflow-aggregator), and JUnit (junit). If missing, find them under Available plugins, install them with their dependencies, and restart Jenkins if requested.

5. Open Manage Jenkins > Nodes > Built-In Node > Configure. Set Number of executors to 0 and save. We will configure a separate build agent together later. No agent, job, repository, or pipeline needs to be created now.

9. Know how to stop and restart

After checking the dashboard, stop Jenkins with Ctrl+C in its terminal. To restart, open a terminal, change to the same jenkins-lab folder using step 5, and repeat both commands in step 7 for your OS. Keeping the same JENKINS_HOME preserves your account and plugins; no repeat wizard is needed.

Keep the entire CloudJenkinsLab folder for the session. Do not delete runtime or upload it: it contains your Jenkins configuration and credentials.

If something fails

Java/Git not found: finish installation and reopen the terminal. Wrong Java: correct PATH to use JDK 21, not only your IDE's Java. Unable to access jarfile: check you are inside jenkins-lab and the download is named jenkins.war, not jenkins.war.zip or jenkins.war.txt.

Browser cannot connect: leave the terminal open and check for startup errors. Port 8080 already in use: do not stop unrelated applications; ask the instructor for help. Download/plugin failure: check Wi-Fi and campus proxy access. If college policy blocks installation, ask IT; do not disable security controls or change machine-wide policies.

Bring the exact error text, your OS, and java/javac version output if you need help. Remove all passwords and secrets from screenshots.

You are ready when...

[ ] Java tools report 21.x and Git reports a version in your terminal.

[ ] You can sign in to the local Jenkins dashboard and the three plugins are installed.

[ ] Built-in executors are 0, and you know how to stop and restart Jenkins.

[ ] Your laptop, charger, editor, browser, and internet access are ready.

That is all for preparation. The project, code changes, and automated build exercise will be introduced during the lecture.